IC frontier campaign — Phase 1 run report

Run. 2026-10-15, sessions 07:09–08:43 UTC. Protocol `PROTOCOL.md` (frozen before the run), specs `phase1/spec-*.json`, parameters `params.json`. Host: repository cloud container, Linux x86_64, 4 vCPU, 15 GiB; `--cpus none --allow-busy`: every record is **L0**, operation counts only, no wall time admitted. The `ecbench` binary was rebuilt during session P32 with analysis-only changes (`phase1/BINARY_NOTE.txt`).

Workloads. Public hash-to-subgroup hold-out targets (seed 20261015200), 16 per curve, 2 measured rounds and 1 warm-up. Koblitz $(a, n) \in \{(0,13), (1,17), (1,19), (1,23), (0,37), (1,43), (1,47), (0,41)\}$; $a = -3$ prime-order curves of 20, 24, 28, 32, 36, 40 bits; PKM ladder curves of 20–28 bits.

Requested and completed. 19 sessions, 4,848 executions, 3,232 measured: 3,178 verified and 54 exhausted (all `ic-shared`: 30 at $n = 41$, 24 at $n = 47$). No timeouts, errors or crashes. Replay-all audits: 19/19 sessions `ok`, every replayed record identical (`phase1/audits/`).

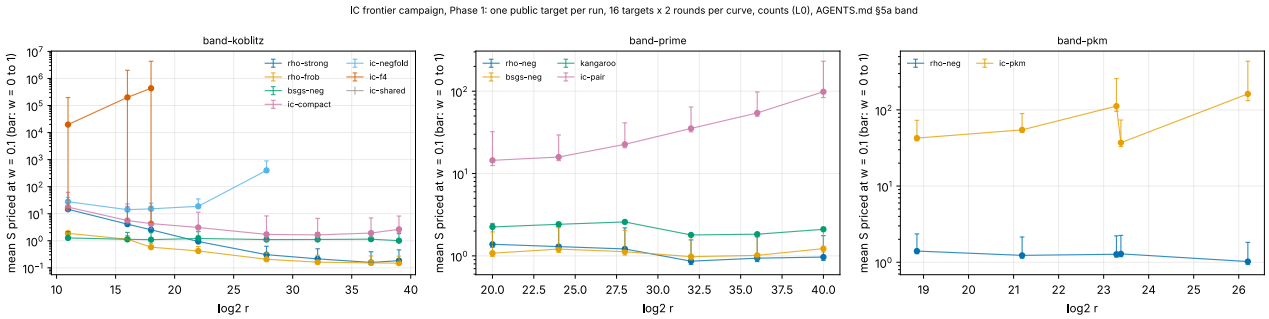


Figure 1: Mean S per arm and curve at $w = 0.1$ GAE per native operation, bar from $w = 0$ to $w = 1$ (`ecbench band`, `analysis/band-*.json`). Open marker: not every run verified.

Observations (the harness's band verdicts, `analysis/band-*.md`). On every Koblitz curve `rho-frob` < `ic-compact` and `rho-frob` < `ic-negfold` (where it runs, $n \leq 37$) at all three weights; the one unresolved IC–rho pair is `ic-compact` vs `rho-strong` at $n = 13$. On every prime and PKM curve `rho-neg` < every IC arm at all three weights. IC/best-rho mean-S ratios: `analysis/ratios_to_rho.tsv`. Exponent fits per arm, tier and weight (`ecbench bound fit --weight [--param-rule]`): `analysis/bounds/` (63 records); no IC arm has a fitted exponent below the matching rho exponent. `ic-shared` has uncounted unpriced categories and is reported at $w = 0$ only.

Raw records. `phase1/sessions/<curve>/records.jsonl`. Interpretation: `README.md` and the paper.